

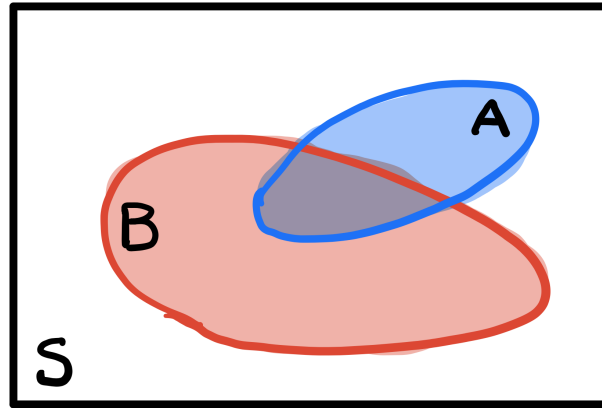


Figure 2.10

DRAFT

2.5 Conditional probability

Conditional probability answers the question “if I acquire knowledge that an event B occurred, how should this news change my assessment of the probability of another event A ?” If I know B has occurred, then I am interested in computing a new probability $P(A | B)$, which is “the probability of A given B .” Contrast this with regular ol’ $P(A)$, which is the *marginal* or *unconditional* probability of A .

Definition 2.3. If $A, B \subseteq S$, then the **conditional probability** of A given B is

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(B) > 0. \quad (2.7)$$

To see what this definition is capturing, consider Figure 2.10. Because we *know* that B occurred, we can ignore everything outside of B . In particular, the only part of A that matters now is the part that intersects with B itself, hence the numerator. But if we are ignoring everything outside of B , then B has effectively become our new sample space, and the map $A \mapsto P(A | B)$ is a new probability measure on this revised space. This is why we divide by $P(B)$ in the definition, so that this new probability measure on the sample space B satisfies the axiom of total measure 1:

$$P(B | B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

Because the map $A \mapsto P(A | B)$ is a probability measure in its own right, it satisfies all the axioms and obeys all the rules (check this!).

2.5.1 The marginal/conditional decomposition

Because set intersection is a commutative operation, we have

$$P(B | A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)}.$$

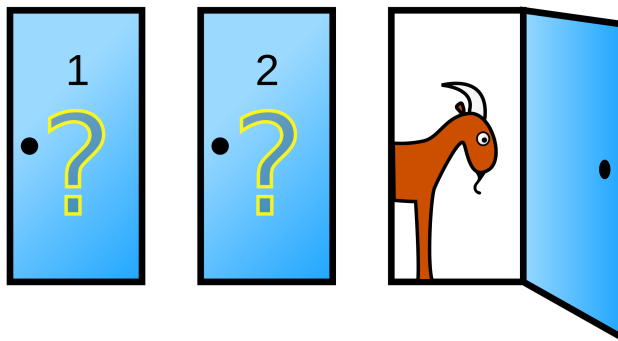


Figure 2.11

This, together with our original definition, provides us with “two identities for the price of one”:

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A).$$

I call this the **marginal/conditional decomposition**. It also goes by the *chain rule of probability* or the *multiplication rule*. In general, the probability of an intersection (a joint probability) can always be rewritten as the product of a marginal probability and a conditional probability. This is one of the most useful ways to compute the probability of an intersection, and one application of this is to rewriting the law of total probability:

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i) = \sum_{i=1}^{\infty} P(A | B_i)P(B_i), \quad (B_i) \text{ is a partition of } S. \quad (2.8)$$

When you apply the law of total probability, you will probably use it in this second form.

2.5.2 Bayes’ theorem

If we take our original definition of conditional probability and rewrite the numerator using a marginal/conditional decomposition, we get Bayes’ theorem:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}. \quad (2.9)$$

This is useful because it is hyper-explicit about the exact relationship between the marginal probability $P(A)$ and the conditional probability $P(A | B)$. This formula tells you exactly how your beliefs about the event A ought to be updated or revised in light of the news that B occurred.

2.5.3 Example: the Monty Hall problem

This is a famous problem in probability, based on the game show *Let’s Make a Deal* hosted by Monty Hall. In the game, a contestant is presented with three doors. Behind one of the doors is the grand prize of a new car, and behind each of the other two doors is a goat (visualized in Figure 2.11). The goal is to ultimately select the door that has the prize behind it. The game proceeds as follows:

- (i) You select one of the doors to open;

- (ii) Instead of opening the door you chose, the host will open one of the two remaining doors that has a goat behind it;
- If the door you first chose has the prize behind it, then the host will select between the two remaining doors with equal probability;
 - If the door you first chose has a goat behind it, then of course the host will select the only other door with a goat behind it;
- (iii) The host will then give you a choice: stick with the original door you selected, or switch to the other remaining one. What should you do?

When most people encounter this problem for the first time, they guess that there is a 50/50 chance that the prize is behind either of the two remaining, unopened doors, but this is false. Let's say without loss of generality that the contestant initially selects Door 1, and then the host opens Door 2 to reveal a goat. Once it is revealed that Door 2 has a goat behind it, Door 3 becomes *conditionally* more likely to conceal the prize. This is because there are circumstances under which the host would be actively avoiding Door 3 because it conceals the prize. If the original choice of Door 1 has a goat behind it, then the host is constrained not to pick the door with the prize inside. The fact that he has not opened Door 3 tips you off to the possibility that he is hiding something, so his revealing Door 2 makes Door 3 more likely, and your best bet is to switch.

In order to see this beyond a shadow of a doubt, we will use Bayes' theorem to compute and compare the following:

$$P(D1 \text{ prize} | D2 \text{ open}) = \frac{P(D2 \text{ open} | D1 \text{ prize})P(D1 \text{ prize})}{P(D2 \text{ open})}$$

$$P(D3 \text{ prize} | D2 \text{ open}) = \frac{P(D2 \text{ open} | D3 \text{ prize})P(D3 \text{ prize})}{P(D2 \text{ open})}.$$

To begin, we know that

$$P(D1 \text{ prize}) = P(D2 \text{ prize}) = P(D3 \text{ prize}) = \frac{1}{3}.$$

Next note that

$$\begin{aligned} P(D2 \text{ open} | D1 \text{ prize}) &= 1/2 \\ P(D2 \text{ open} | D2 \text{ prize}) &= 0 \\ P(D2 \text{ open} | D3 \text{ prize}) &= 1. \end{aligned}$$

The law of total probability says

$$\begin{aligned} P(D2 \text{ open}) &= P(D2 \text{ open} | D1 \text{ prize})P(D1 \text{ prize}) + \\ &\quad P(D2 \text{ open} | D2 \text{ prize})P(D2 \text{ prize}) + \\ &\quad P(D2 \text{ open} | D3 \text{ prize})P(D3 \text{ prize}) \\ &= \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} \\ &= 1/2, \end{aligned}$$

	$D = +$	$D = -$
$T = +$	True Positive (TP)	False Positive (FP)
$T = -$	False Negative (FN)	True Negative (TN)

Table 2.6: Contingency table of test results vs. disease status.

so we're ready to go:

$$\begin{aligned}
 P(D1 \text{ prize} \mid D2 \text{ open}) &= \frac{P(D2 \text{ open} \mid D1 \text{ prize})P(D1 \text{ prize})}{P(D2 \text{ open})} \\
 &= \frac{(1/2)(1/3)}{1/2} \\
 &= 1/3
 \end{aligned}$$

$$\begin{aligned}
 P(D3 \text{ prize} \mid D2 \text{ open}) &= \frac{P(D2 \text{ open} \mid D3 \text{ prize})P(D3 \text{ prize})}{P(D2 \text{ open})} \\
 &= \frac{1 \cdot (1/3)}{1/2} \\
 &= 2/3.
 \end{aligned}$$

You are more likely to win if you switch.

2.5.4 Example: disease testing

Consider a diagnostic test for a disease. Tests work pretty well in a lot of cases, but they are not infallible. A test could say that you have a disease when you do not (false positive), or it could fail to detect a disease that you in fact have (false negative). Let's be more concrete. Let D denote someone's true disease status; they either have the disease (+) or they don't (-). Let T denote the result of someone's test; the test either says they have the disease (+), or it says they don't (-). We will assume that these are the only two attributes a person possesses, so our little model is ignoring important things like symptoms, genetics, medical history, lifestyle, etc. Table 2.6 displays the sample space of this phenomenon.

There are several probabilities associated with this setup, and we give them special names:

- **prevalence:** $P(D = +)$ is the marginal probability of contracting the disease. If you had no other information, this is the baseline likelihood that a randomly selected member of the population is sick;
- **false negative rate (FNR):** $P(T = - \mid D = +)$ is the probability of the test saying you don't have the disease when you do;
- **false positive rate (FPR):** $P(T = + \mid D = -)$ is the probability of the test saying you have the disease when you do not;

- **sensitivity (true positive rate):** $P(T = + | D = +) = 1 - P(T = - | D = +)$ is the probability of the test correctly detecting that you have the disease;
- **specificity (true negative rate):** $P(T = - | D = -) = 1 - P(T = + | D = -)$ is the probability of the test correctly detecting that you *do not* have the disease.

We would love for the sensitivity and the specificity of a test to both be high, but there is a trade-off between them. Up to a certain point, you cannot improve sensitivity without making specificity worse, and vice versa. You can always make either the sensitivity or specificity perfect, but it will be at the expense of the other:

- Imagine a test that automatically returns (+) for everyone that takes it. In that case, everyone with the disease will be correctly diagnosed, and so the sensitivity $P(T = + | D = +) = 1$ is perfect. But there are a lot of healthy people being diagnosed with a disease they don't have, and so the specificity $P(T = - | D = -) = 0$ is perfectly wretched;
- Conversely, imagine a test that automatically returns (-) for everyone that takes it. So everyone with the disease is being told they don't have it, and the sensitivity $P(T = + | D = +) = 0$ sucks. But everyone without the disease is correctly being told they're healthy, and so the specificity $P(T = - | D = -) = 1$ is now perfect.

Here's another analogy to drive it home. Think of D as someone's guilt (+) or innocence (-) of committing a crime, and think of T as the verdict of guilt (+) or innocence (-) rendered by a judge.

- If the judge wants to trivially max out sensitivity, she should just declare everyone before her guilty and send everyone to jail. In this way, no criminals are walking the street, but now you have a lot of innocent people locked up;
- If the judge wants to trivially max out specificity, she could let everyone go free. Now no innocent people are jailed, but you have criminals on the loose.

We seek the “Goldilocks sweet spot” between these two absurd extremes, but there's a real tradeoff. A “perfect test” would have sensitivity and specificity both equal to 1, but if the outcomes are truly random (meaning not perfectly predictable), then a perfect test is not possible. You'll always make some mistakes some of the time.

Example 2.12. A new disease has reached our shores. It is caused by a rare bacteria called *Salmonella Fitzgerald*. Those infected with *Salmonella Fitzgerald* are randomly stricken with an irresistible urge to scat sing. This is a rare disease, and the overall prevalence in the population is only $P(D = +) = 0.01$. During a routine checkup, Dr. Vinnie Boombatz administers a diagnostic test for *Salmonella Fitzgerald* that has a true positive rate (sensitivity) of $P(T = + | D = +) = 0.9$ and a false positive rate of $P(T = + | D = -) = 0.09$. The test comes back positive, but it's clearly not a perfect test, so the patient asks the good doctor what is the revised probability that they have the disease, given this new information. That is, what is $P(D = + | T = +)$? This is a job for Bayes' theorem:

$$P(D = + | T = +) = \frac{P(T = + | D = +)P(D = +)}{P(T = +)}.$$

We are given the prevalence $P(D = +) = 0.01$ and the sensitivity $P(T = + | D = +) = 0.9$, so it remains to calculate $P(T = +)$. This is a job for the law of total probability. Table 2.6 is our sample space, and we will partition it according to the value of D . So the law of total probability¹ allows us to write:

$$\begin{aligned}
 P(T = +) &= P(T = + \cap D = -) + P(T = + \cap D = +) \\
 &= P(T = + | D = -)P(D = -) + P(T = + | D = +)P(D = +) \\
 &= P(T = + | D = -)[1 - P(D = +)] + P(T = + | D = +)P(D = +) \\
 &= 0.09(1 - 0.01) + 0.9 \cdot 0.01 \\
 &= 0.0981.
 \end{aligned}$$

So we have

$$P(D = + | T = +) = \frac{P(T = + | D = +)P(D = +)}{P(T = +)} = \frac{0.9 \cdot 0.01}{0.0981} \approx 0.0917.$$

So, before being tested, the probability the patient had the disease was 0.01. After testing positive, we revise this up to 0.09. Testing positive is certainly not good news, but it's also not a "death sentence." Why? Because, even though the test seems pretty good (sensitivity of 0.9 and specificity of 0.91), the disease is just so rare (prevalence of 0.01) that it's unlikely *anyone* has it, regardless their test result.

Figure 2.12 displays the value of $P(D = + | T = +)$ as a function of the prevalence for various combinations of TPR and TNR. The upper right panel corresponds to the present example. Across the board, we see that the revised probability is always greater (the curve is above the 45 degree line), but *how much* greater will depend. The rare disease we considered places us in the lower left corner where testing positive does not increase the likelihood of disease as much as it would for a more common ailment.

By looking across the other panels in Figure 2.12, we can see how the error rates affect the informativeness of the test. When $TPR = TNR = 0.5$, then $P(D = + | T = +) = P(D = +)$. The disease status and the test result are independent, and testing positive teaches us nothing. You are just as likely to have the disease as you were before you took the test. The closer TPR and TNR each are to 1, the larger the gap between the marginal and conditional probability. If TPR and TNR are close to 0, the test is *anti-informative*; testing positive makes you *less* likely to have the disease.

Remark 2.8. That example was admittedly a tad silly, but the numbers are taken from a *BBC News* article entitled "[Do doctors understand test results?](#)" The question was posed to a group of 160 physicians, and only 21% answered correctly. Almost half said that $P(D = + | T = +)$ was 0.9, which is the test's sensitivity. These doctors are committing something called the **base rate fallacy**. They are ignoring the prevalence (the *base rate*) of the disease when analyzing the problem. They are also neglecting the fact that the test commits two kinds of errors: false negatives and false positives.

¹In the notation of Section 2.3.4, the event of interest is $A = \{T = +\}$ and our partition of the sample space consists of $B_1 = \{D = -\}$ and $B_2 = \{D = +\}$.

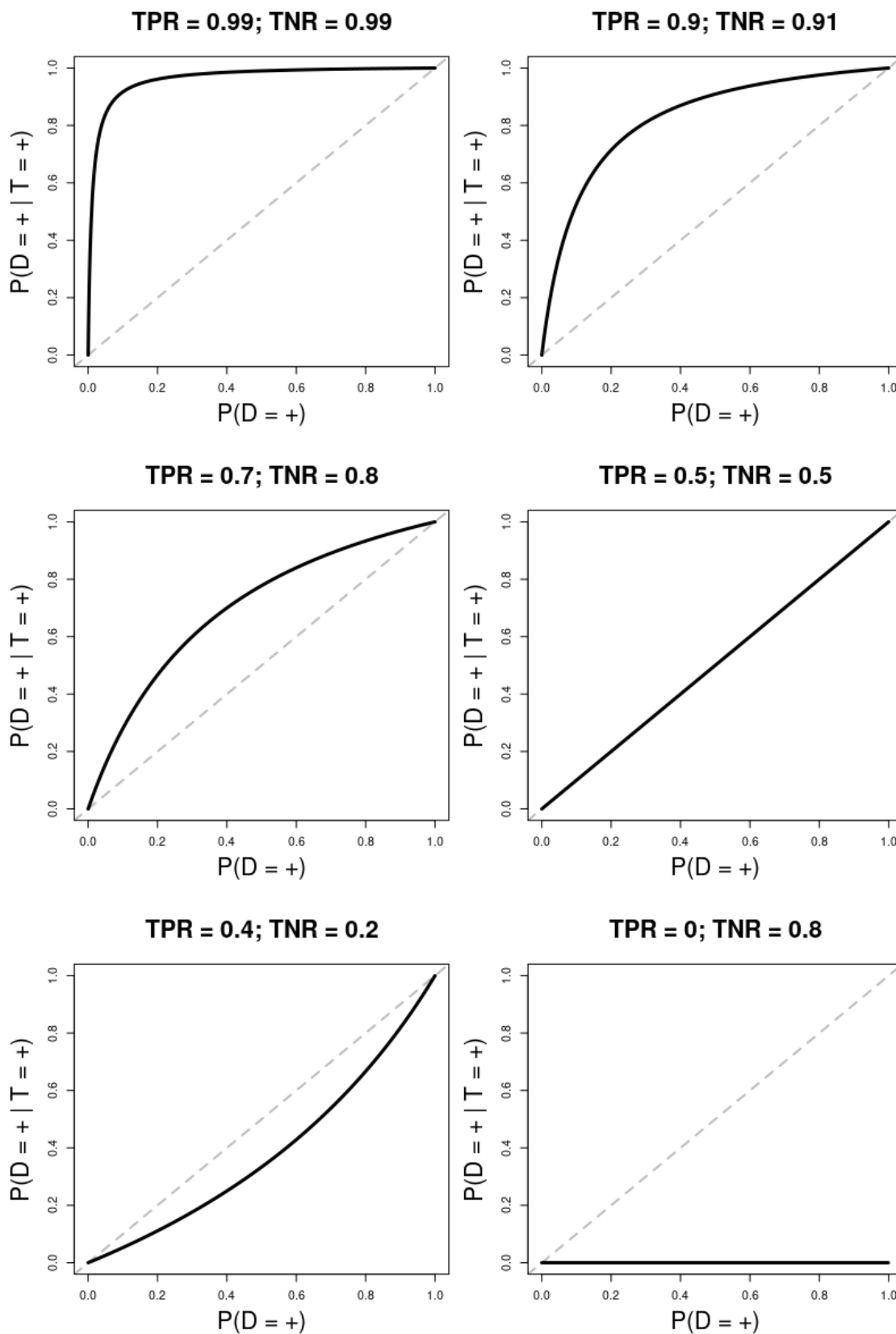


Figure 2.12